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Teo densidad induce metrica

Teorema 1. *Sea* $\mu\colon \Omega \subset \mathbb{R}^n \to \mathbb{R}$ *continua y positiva con* Ω *conexo, definimos*

$$\forall\,\Gamma \text{ camino inyectivo en } \Omega : \ell_\mu(\Gamma) = \int_a^b \mu(\Gamma(t))\, \|\Gamma'(t)\| \,\mathrm{d}t.$$

Entonces, la aplicación $d\colon \Omega \times \Omega \to [0, \infty)$ *dada por*

$$\forall x, y \in \Omega : d(x, y) = \inf \left\{ \ell_\mu(\Gamma) : \Gamma \text{ camino inyectivo en } \Omega \text{ entre } x \text{ e } y \right\},$$

es una métrica en Ω*. A la aplicación* μ *se le llama la* ***densidad métrica*** *de* d*.*

Referenciado en

- `densidad-metrica-inducida`

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